

Learning Bigram Language Models

- The parameters of a Markov LM can be represented as a stochastic, positive-valued matrix (*transition matrix*)
 - Sometimes the distribution of the first symbol is represented separately, but we can think about the sequences as always starting with n instances of a special START symbol

$$A_{ij} = P(x_m = w_j | x_{m-1} = w_i)$$

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- A maximum likelihood estimator for the transition matrix is obtained by counting:

$$q_{ML}(x_m = w_j | x_{m-1} = w_i) = \frac{\text{count}(w_i, w_j)}{\sum_{j'} \text{count}(w_i, w_{j'})}$$

Learning Higher-order Language Models

- A maximum likelihood estimator for a k -th order language model is:

$$q_{ML}(x_m = w_j | x_{m-1} = w_{i1}, x_{m-2} = w_{i2}, \dots, x_{m-k} = w_{ik}) = \frac{\text{count}(w_{ik}, \dots, w_{i1}, w_j)}{\sum_{j'} \text{count}(w_{ik}, \dots, w_{i1}, w_{j'})}$$